Now select the baud rate as '9600,' data bits as '8,' parity as 'none,' stop bits as '1,' flow control as 'Xon/Xoff' and click 'ok' (refer Fig. 6).

The microcontroller counts the received pulses and sends the pulse count to the hyper terminal and LCD. You will see a message on the screen as shown in Fig. 8. Simultaneously, the pulse count appears on the LCD.

Software

The software for this project is given at the end of this article. It is written in 'C' language and compiled using Keil µVision4 compiler. Burn the generated hex code into the microcontroller using a suitable programmer.

In the software, first enable timer and serial interrupts. Then configure the timer by mode and reload value



Fig. 6: Hyperterminal window for selection of baud rate, data bits, parity, stop bits and flow control



Fig. 7: Count on hyperterminal screen

to give the correct speed (9600 bits per second). SCON gives the necessary UART signal and your data is sent serially. When port pin P3.5 (T1) goes low, the internal counter increases by one. This count is displayed on the LCD as well as hyper terminal of the PC.

EFY note. The source code of this project has been included in this month's EFY-CD and is also available on www.efymag.com website.

COUNTER.C

```
#include<reg51.h>
#define LCD P1
unsigned int Count,d1,d2,d3,d4,x1,x2,x3,x4;
sbit Pulse      = P3^5;
sbit LED  = P3^7;
sbit rs         = P3^4;
sbit rw         = P3^3;
sbit en         = P3^2;
void LCDInit ();
void lcdcmd          (unsigned char);
void lcdwrt          (unsigned char);
void Delay           (unsigned int);
void Convert (unsigned int);
void SerialTx (unsigned char);
void main(void)
{
        unsigned char z,Display;
        unsigned  char  code  str1[]="
PULSE COUNTER  ";
        unsigned  char  code  str2[]="
Design By      ";
        unsigned  char  code  str3[]="
HIMANSHU       ";
        unsigned  char  code  str4[]="
SHARMA         ";
        unsigned char code str5[]="PULSE:
0000           ";
        P0=0xFF;
        P1=0xFF;
        P2=0xFF;
        P3=0xFF;
        IE=0x00;
        Count=0x00;
//////SERIALIZATION DATA//////
        TMOD        = 0x20;
        TH1                 = 0xFD;
        SCON        = 0x50;
        TR1                 = 1;
        LED                 = 0x00;
        LCDInit();
        for(z=0;z<16;z++)
        {
                Display=str1[z];
                lcdwrt(Display);
                Delay(1);
                SerialTx(Display);
                Delay(1);
        }
        SerialTx('\r');
        SerialTx('\n');
        lcdcmd(0xC0);
        for(z=0;z<16;z++)
        {
                Display=str2[z];
                lcdwrt(Display);
                Delay(1);
                SerialTx(Display);
                Delay(1);
        }
        Delay(200);
        SerialTx('\r');
        SerialTx('\n');
        lcdcmd(0x01);
        Delay(1);
for(z=0;z<16;z++)
        {
                Display=str3[z];
                lcdwrt(Display);
                Delay(1);
                SerialTx(Display);
                Delay(1);
        }
        SerialTx('\r');
        SerialTx('\n');
        lcdcmd(0xC0);
        for(z=0;z<16;z++)
        {
                Display=str4[z];
                lcdwrt(Display);
                Delay(1);
                SerialTx(Display);
                Delay(1);
        }
        Delay(200);
        SerialTx('\r');
        SerialTx('\n');
        lcdcmd(0x01);
        Delay(1);
        for(z=0;z<16;z++)
        {
                Display=str5[z];
                lcdwrt(Display);
                Delay(1);
                SerialTx(Display);
                Delay(1);
        }
        Delay(70);
        SerialTx('\r');
        SerialTx('\n');
        while(1)
        {
                LED=i;
                Delay(30);
                if(Pulse==0)
                {

Count=Count+1;
Convert(Count);
if(Count==9999)
                                Count=0x00;
                }
        }
}
void Convert(unsigned int value)
{
        x1 = value / 10;
        d1 = value % 10;
        x2 = x1 / 10;
        d2 = x1 % 10;
        x3 = x2 / 10;
        d3 = x2 % 10;
        d4 = x3 % 10;
        d1=d1|0x30;
        d2=d2|0x30;
        d3=d3|0x30;
        d4=d4|0x30;
        lcdcmd(0xC7);
        lcdwrt(d4);
        lcdwrt(d3);
        lcdwrt(d2);
        lcdwrt(d1);
        Delay(1);
        SerialTx(d4);
        SerialTx(d3);
        SerialTx(d2);
        SerialTx(d1);
        Delay(1);
        SerialTx('\r');
        SerialTx('\n');
        LED=0;
        Delay(30);
}
void LCDInit()
{
        lcdcmd(0x38);
        Delay(10);
        lcdcmd(0x0E);
        Delay(10);
        lcdcmd(0x01);
        Delay(10);
        lcdcmd(0x0C);
        Delay(10);
        lcdcmd(0x80);
        Delay(10);
        return;
}
void lcdcmd(unsigned char value)
{
        LCD=value;
        rs=0;
        rw=0;
        en=1;
        Delay(1);
        en=0;
        return;
}
void lcdwrt(unsigned char value)
{
        LCD=value;
        rs=1;
        rw=0;
        en=1;
        Delay(1);
        en=0;
        return;
}
void SerialTx(unsigned char value)
{
        SBUF = value;
        while(TI==0);
        TI = 0;
}
void Delay(unsigned int x)
{
        unsigned int i,j;
    for (i=0;i<=x;i++)
    for (j=0;j<=500;j++);
}
```